fiTQun reconstruction workflow

From PMT hits to subevents, one-ring and multi-ring likelihood fits

Measured quantities and Hypothesis X

Measured quantities

- For each PMT: measured charge q_i, raw hit time t_i^{raw}, PMT position $\boldsymbol{R}_i$ and orientation.
- Here t_i^{raw} is taken as already expressed relative to the event trigger.

Unknown reconstructed quantities for a one-ring hypothesis

$$X = \{t, x, y, z, \theta, \phi, p\}$$

- Vertex time and position: $(t, \vec{x})$.
- Direction: $\hat{\boldsymbol{d}}(\theta, \phi)$.
- Momentum: p.

Chronological overview

1. **Prompt vertex pre-fit for subeventing**
   - Use prompt hits in $[-100, +400]$ ns around the trigger.
   - Estimate a rough vertex $\boldsymbol{x}_{\mathrm{vtx}}^{\mathrm{pre}}$.
2. **Peak finder / subevent algorithm**
   - Fix $\boldsymbol{x} = \boldsymbol{x}_{\mathrm{vtx}}^{\mathrm{pre}}$.
   - Scan goodness $G(\boldsymbol{x}_{\mathrm{vtx}}^{\mathrm{pre}}, t)$ in time.
   - Identify time peaks and build subevent windows.
3. **Single-ring fit for each subevent**
   - For each particle hypothesis: e, μ, π^+, etc.
   - Maximize the charge–time likelihood over X.
4. **Special one-ring hypotheses**
   - In-gate decay-electron hit masking.
   - Upstream-track π^+ and two-photon π^0 fits.
5. **Multi-ring reconstruction**
   - Applied to the primary subevent.
   - Add and refit several simultaneous Cherenkov rings.

1. Vertex pre-fit: what it does

Purpose

- Fast, rough estimate of a point-like light source.
- Uses PMT timing information only.

Residual time for a point-like source

$$T^i_{\mathrm{res}}(\boldsymbol{x}, t) = t_i^{\mathrm{raw}} - t - \frac{|\boldsymbol{R}_i - \boldsymbol{x}|}{c_n}$$

Goodness function

$$G(\boldsymbol{x}, t) = \sum_{i \in \mathrm{hit}} \exp\left[-\frac{1}{2}\left(\frac{T^i_{\mathrm{res}}}{\sigma}\right)^2\right]$$

Physical meaning

- Correct vertex/time $\Rightarrow$ direct-light hits have $T_{\mathrm{res}} \simeq 0$.
- Many terms close to 1 $\Rightarrow$ large G.
- Wrong vertex/time $\Rightarrow$ residuals spread out $\Rightarrow$ lower G.

1. Vertex pre-fit: role in the subevent algorithm

- The vertex pre-fitter is a general fast algorithm.
- In the peak finder, it is applied specifically to **prompt hits**: $[-100, +400]$ ns around event trigger
- Output: rough prompt vertex $\boldsymbol{x}_{\rm vtx}^{\rm pre}$.

Why this is needed

- The peak finder wants to locate activity in time.
- PMT hit times must first be corrected for photon time of flight.
- A rough spatial reference is enough:

$$T_{\rm res}^{i}(t) = t_i^{\rm raw} - t - \frac{|\boldsymbol{R}_i - \boldsymbol{x}_{\rm vtx}^{\rm pre}|}{c_n}.$$

- **Approximation**: the particle emits all the cherenkov photons at its vertex.
- $\boldsymbol{x}_{\rm vtx}^{\rm pre}$ is a seed / timing reference.
- The final single-ring likelihood fit later refits vertex, time, direction and momentum.

2. Peak finder: time scan

Fixed quantity

$$\boldsymbol{x} = \boldsymbol{x}_{\rm vtx}^{\rm pre}$$

Scanned quantity

$$G(t) \equiv G(\boldsymbol{x}_{\rm vtx}^{\rm pre}, t)$$

Practical scan

- Step size:

$$\Delta t = 8 \text{ ns}.$$

- Start: about 300 ns before the trigger.
- Hits used for the scan: [-200, 15000] ns around the event trigger.
- Width used in the goodness during this scan:

$$\sigma = 6.3 \text{ ns}.$$

Why?

- Each physical activity creates a cluster of residual times.
- A subevent appears as a peak in $G(t)$.

2. Peak finder: local maxima

Goodness scan points

$$\{t_k,\ G(t_k)\}, \qquad t_{k+1} - t_k = 8\ \mathrm{ns}$$

Local maximum condition

$$G(t_k) > G(t_{k-1}), \qquad G(t_k) > G(t_{k+1})$$

Basic noise rejection

$$G(t_k) > \eta, \qquad \eta = 9.$$

Set of candidate local maxima

$$M = \{t_k :\ G(t_k) > G(t_{k\pm1}),\ G(t_k) > \eta\}\,.$$

Why local maxima are not enough

- One physical subevent may produce several nearby maxima.
- Reason: TOF correction assumes a point source at $\boldsymbol{x}_{\mathrm{vtx}}^{\mathrm{pre}}$.
- Real Cherenkov emission is extended along the particle track.

2. Peak finder: dynamic threshold $F(t)$

Threshold curve

$$F(t) := 0.25 \max_{i \in M} \{ G(\boldsymbol{x}_{\mathrm{vtx}}^{\mathrm{pre}}, t_i)\, f(t - t_i) \} + \eta$$

Shape function

$$f(\tau) = \frac{1}{1 + (\tau/\gamma)^2}$$

Asymmetric time scale

$$\gamma = \begin{cases} 25 \text{ ns}, & \tau < 0, \\ 70 \text{ ns}, & \tau > 0. \end{cases}$$

Meaning

- $F(t)$ is high near already strong local maxima.
- $F(t)$ decreases away from them.
- The offset $\eta = 9$ suppresses low-goodness noise peaks.
- The longer tail for $\tau > 0$ is conservative for delayed bumps after a peak.

2. Peak finder: accepting the first and later peaks

First accepted peak

- Scan forward in time.
- The first subevent peak is the first local maximum $t_i \in M$ satisfying
$$G(t_i) > F(t_i).$$

Further peaks: double-threshold condition

- After an accepted peak, continue scanning to later times.
- A later local maximum above $F(t)$ is **not accepted immediately**.
- Before accepting a new peak, the scan must have gone below
$$0.6F(t)$$
after the previous accepted peak.

Physical reason

- Prevent double-counting shoulders / secondary bumps of the same subevent.
- Accept a new peak only after the goodness valley becomes sufficiently deep.

2. Defining subevent time windows

For each accepted peak time t_{peak}

$$T^i_{\text{res}} = t^{\text{raw}}_i - t_{\text{peak}} - \frac{|\boldsymbol{R}_i - \boldsymbol{x}^{\text{pre}}_{\text{vtx}}|}{c_n}$$

Associated hits

$$-180 \text{ ns} < T^i_{\text{res}} < 800 \text{ ns}$$

Window edges

- Among the hits satisfying the residual-time cut:

$$t_{\text{start}} = \min(t^{\text{raw}}_i), \qquad t_{\text{end}} = \max(t^{\text{raw}}_i).$$

- The subevent window is

$$[t_{\text{start}}, t_{\text{end}}].$$

Overlapping windows

- If two windows overlap, they are merged.

2. What the time window means for the likelihood

Inside one subevent window

- A PMT can contribute at most one hit to the subevent likelihood.
- Hit PMTs: PMTs with a selected hit **inside the window**.
- Unhit PMTs: PMTs without a selected hit inside the window.

Crucial convention

- Hits outside the current time window are **not used as hits** for this subevent.
- For this subevent reconstruction, they are treated as unhit in the likelihood.

Why this matters

- Separates prompt activity from delayed activity.
- Example:
  - prompt muon subevent;
  - delayed Michel electron subevent.
- Avoids fitting both with one single time PDF.

3. Single-ring likelihood: hypothesis and variables

One-ring hypothesis

$$\mathcal{H} = \{\text{one particle of type } a\}, \qquad a \in \{e, \mu, \pi^+, \ldots\}$$

Track parameters

$$X = \{t, x, y, z, \theta, \phi, p\}$$

Likelihood

$$L_a(X) = \prod_{j \in \mathrm{unhit}} P_j(\mathrm{unhit}|\mu_j) \prod_{i \in \mathrm{hit}} [1 - P_i(\mathrm{unhit}|\mu_i)]\, f_q(q_i|\mu_i)\, f_t(t_i|X)$$

Fit result for each particle hypothesis

$$\hat{X}_a = \arg\max_X L_a(X)$$

PID idea

$$\Delta \ln L_{e/\mu} = \ln L_e(\hat{X}_e) - \ln L_\mu(\hat{X}_\mu)$$

3. What is computed in the likelihood?

For each trial X during minimization

1. Compute expected charge at each PMT:
$$\mu_i(X) = \mu_i^{\mathrm{dir}}(X) + \mu_i^{\mathrm{sct}}(X).$$
2. Evaluate unhit probability:
$$P_i(\mathrm{unhit}|\mu_i).$$
3. Evaluate charge PDF:
$$f_q(q_i|\mu_i).$$
4. Evaluate time PDF:
$$f_t(t_i|X).$$
5. Sum contributions to $-\ln L_a(X)$.

Minimization

- MINUIT / SIMPLEX-like minimization of $-\ln L_a(X)$.
- Vertex, time, direction and momentum are varied together after seeding.

4. Predicted charge: direct light

Formal expression

$$\mu^{\mathrm{dir}} = \Phi(p) \int ds \, g(p, s, \cos\theta) \, J(s, \eta)$$

Ingredients

- s: position along the track.
- $g(p, s, \cos\theta)$: Cherenkov emission profile.
- $J(s, \eta)$: optical/geometrical response.
  - solid angle;
  - water attenuation;
  - PMT angular acceptance;
  - relative geometry between track element and PMT.
- $\Phi(p)$: total light normalization / mean photon yield factor.

Physical picture

- Integrate light emitted along the particle track.
- Weight by probability that photons reach and are accepted by the PMT.

4. Direct light: tuning vs on-the-fly

Precomputed / tuned with MC

- Cherenkov profiles $g(p, s, \cos\theta)$.
- Cherenkov integrals over binned values of momentum and geometry.
- PMT angular response tables.
- Optical response parameters.

Parabolic approximation

$$J(s) \simeq j_0 + j_1 s + j_2 s^2 \Rightarrow \mu^{\mathrm{dir}} \simeq \Phi(p)\,(I_0 j_0 + I_1 j_1 + I_2 j_2)$$

$$I_n(r_0, \theta_0, p) = \int ds\; g(p, s, \cos\theta)\, s^n$$

On-the-fly during the fit

- Geometry between current trial track X and each PMT.
- Coefficients j_0, j_1, j_2 from evaluating $J(s)$ at 3 representative points(vertex, position where the 90% average nr of photons is emitted, mid-point between these two points).
- Interpolation of precomputed integrals I_n evaluated at the found values of p, r, θ.

5. Predicted charge: scattered / indirect light

Idea

- Indirect light = photons arriving after scattering/reflection.
- Model as an imaginary isotropic light source along the same trajectory.
- Correct the isotropic source using a scattering table $A(s)$.

Scattering table

$$A(s) = \frac{d\mu_{\mathrm{sct}}}{d\mu_{\mathrm{iso,dir}}}$$

Approximation

- Cherenkov angle assumed independent of momentum.
- Equivalent to using $\beta \simeq 1$ for the table construction.
- Momentum dependence factors mostly outside the table.

Parabolic approximation including $A(s)$

$$J(s)A(s) \simeq k_0 + k_1 s + k_2 s^2$$

5. Scattering table: MC construction vs fit evaluation

Precomputed with WCSim / MC

- Generate point-like Cherenkov sources: low-momentum electrons, $\beta \simeq 1$, multiple scattering turned off.
- For photons arriving at PMTs:
  - classify direct photons;
  - classify scattered/reflected photons.
- Fill histograms in detector/track/PMT geometry variables.
- Compute ratio:

$$A = \frac{h_{\mathrm{indirect}}}{h_{\mathrm{direct/isotropic}}}.$$

On-the-fly during the fit

- For the trial track and PMT, compute the six geometry variables (that define the geometry between particle direction and position and the PMT position).
- Interpolate A in the scattering table.
- Compute k_0, k_1, k_2 and then μ_i^{sct}.

6. Unhit probability

Ideal Poisson picture

$$P(0|\mu) = e^{-\mu}$$

fiTQun empirical correction

$$P(\mathrm{unhit}|\mu) \simeq \left(1 + a_1\mu + a_2\mu^2 + a_3\mu^3\right) e^{-\mu}$$

Meaning

- μ: predicted mean charge / mean p.e. at that PMT.
- Electronics threshold and PMT response modify the pure Poisson expectation.
- Coefficients a_i absorb detector effects.

Precomputed / tuned

- Coefficients a_i from detector simulation / calibration tuning.

On-the-fly

- Evaluate the polynomial at the current $\mu_i(X)$.

7. Charge likelihood $f_q(q|\mu)$

Given predicted charge μ,what is the probability to observe q?

MC construction

- Generate true p.e. multiplicity: $n_{\mathrm{pe}} \sim \mathrm{Poisson}(\mu)$.
- Simulate PMT/electronics response.
- Obtain q charge distributions for different μ values.
- Fit / smooth the resulting functions.

On-the-fly

- Input measured charge q_i.
- Interpolate fit parameters as a function of q_i.
- Evaluate the resulting function at the current $\mu_i(X)$.

Important separation

- $\mu_i(X)$: light production and propagation model.
- $f_q(q_i|\mu_i)$: PMT/electronics charge response model.

8. Time likelihood: residual time in the ring fit

In the likelihood fit, photons are not assumed emitted at the vertex

- Approximation: photons emitted at the midpoint of the track.

Residual time

$$t^i_{\mathrm{res}} = t_i^{\mathrm{raw}} - t - \frac{s_{\mathrm{mid}}}{c} - \frac{\left|\boldsymbol{R}_i - \boldsymbol{x} - s_{\mathrm{mid}}\hat{\boldsymbol{d}}\right|}{c_n}$$

Definitions

- s_{mid}: half of effective track length.
- $\boldsymbol{x}$: fitted vertex.
- $\hat{\boldsymbol{d}}$: fitted direction.
- t: fitted vertex time.

Why midpoint?

- Better than point-like emission at the vertex.
- Still simple enough for fast likelihood evaluation.
- Approximation worsens for longer tracks.

8. Direct-light time PDF

Direct-light PDF

$$f_t^{\mathrm{dir}}(t_{\mathrm{res}}|\mu_{\mathrm{dir}}, p) \approx \mathcal{N}\left(t_{\mathrm{res}};\ m(\mu_{\mathrm{dir}}, p), \sigma_t(\mu_{\mathrm{dir}}, p)\right)$$

Dependence on μ_{dir}

- PMT records the first arriving photon.
- More expected photons $\Rightarrow$ earlier / narrower first-photon time distribution.
- Hence time PDF depends on predicted direct charge.

Dependence on p

- Higher momentum $\Rightarrow$ longer track.
- Midpoint approximation less exact.
- Residual time distribution becomes broader / shifted.

Goal

- Model the distribution of the hit time residual for direct Cherenkov light:
- Need a fast way to obtain the Gaussian mean and width for any trial value of μ_{dir} and p.

8. Direct-light time PDF: tuning procedure

Precomputed from particle-gun MC

- Generate single-particle MC samples at fixed momenta p_k.
- For each PMT hit, compute t_{res} and μ_{dir};
- Fill 2D histograms: $H_k(t_{\mathrm{res}}, \log \mu_{\mathrm{dir}})$ separately for each generated momentum p_k.
- In each $\log \mu_{\mathrm{dir}}$ bin, fit the t_{res} distribution with a Gaussian getting one mean and one width for each $\log \mu_{\mathrm{dir}}$ bin.
- Fit the Gaussian parameters as polynomials in $\log \mu_{\mathrm{dir}}$:

$$\mu_k^{\mathrm{Gauss}}(\log \mu_{\mathrm{dir}}) = \sum_{n=0}^{6} a_n(p_k)\,[\log \mu_{\mathrm{dir}}]^n, \quad \sigma_k^{\mathrm{Gauss}}(\log \mu_{\mathrm{dir}}) = \sum_{n=0}^{6} b_n(p_k)\,[\log \mu_{\mathrm{dir}}]^n.$$

- Fit each polynomial coefficient as a smooth function of momentum:

$$a_n(p) = \sum_{r=0}^{N_p} \alpha_{n,r}\, p^r, \qquad b_n(p) = \sum_{r=0}^{N_p} \beta_{n,r}\, p^r.$$

- Store these polynomial parameterizations in the time-PDF tuning files.

8. Direct-light time PDF: used during the fit

On-the-fly, for each trial track hypothesis

$$X = \{t_{\mathrm{vtx}}, \mathrm{x}_{\mathrm{vtx}}, \theta, \phi, p\}$$

1. Compute the predicted direct charge at PMT i: $\mu_{\mathrm{dir},i}(X)$.
2. Use the trial momentum p to evaluate the stored coefficient functions: $a_n(p), b_n(p)$.
3. Reconstruct the Gaussian parameters:

$$\mu_i^{\mathrm{Gauss}} = \sum_{n=0}^{6} a_n(p)\,[\log \mu_{\mathrm{dir},i}]^n, \quad \sigma_i^{\mathrm{Gauss}} = \sum_{n=0}^{6} b_n(p)\,[\log \mu_{\mathrm{dir},i}]^n.$$

4. Evaluate the direct-light time likelihood:

$$f_{\mathrm{dir}}(t_{\mathrm{res},i} \mid \mu_{\mathrm{dir},i}, p) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left[-\frac{(t_{\mathrm{res},i} - \mu_i)^2}{2\sigma_i^2}\right].$$

- The MC histograms and polynomial coefficients are precomputed.
- During the fit, fiTQun only evaluates/interpolates the stored functions.
- This avoids building a new time-residual distribution for every tested X.

Indirect-light time PDF

$$f_t^{\mathrm{sct}}(t_{\mathrm{res}}) = \frac{1}{\sqrt{\pi/2}\,\sigma + 2\gamma} \begin{cases} \exp\left(-\dfrac{\tau^2}{2\sigma^2}\right), & \tau < 0, \\ \left(\dfrac{\tau}{\gamma} + 1\right)\exp\left(-\dfrac{\tau}{\gamma}\right), & \tau > 0, \end{cases}$$

$$\tau = t_{\mathrm{res}} - 5 \text{ ns}, \qquad \sigma = 8 \text{ ns}, \qquad \gamma = 25 \text{ ns}.$$

Meaning

- Empirical shape.
- Long positive tail: delayed scattered/reflected photons.

Merging direct and indirect contributions

$$f_t(t_{\mathrm{res},i}) = w_i f_t^{\mathrm{dir}}(t_{\mathrm{res},i}) + (1 - w_i) f_t^{\mathrm{sct}}(t_{\mathrm{res},i}), \quad w_i \equiv \frac{1 - e^{-\mu_{\mathrm{dir},i}}}{1 - e^{-\mu_{\mathrm{dir},i}} e^{-\mu_{\mathrm{sct},i}}}.$$

9. Seeding the one-ring minimization

The full likelihood fit needs seeds for $X = \{t, x, y, z, \theta, \phi, p\}$

- Vertex / time seed:
  - from pre-fit and subevent peak/window information.
- Direction seed:
  - scan likelihood over 400 equally spaced points on the sphere;
  - choose the best candidate direction.
- Momentum seed:
  - rough estimate from total charge;
  - then one-dimensional momentum scan.
- Trajectory is not arbitrary: determined by $(\boldsymbol{x}, t, \hat{\boldsymbol{d}}, p)$ and the particle hypothesis.

Then

- Minimize $-\ln L_a(X)$ with respect to all track parameters.
- Repeat for each particle hypothesis.

10. In-gate decay-electron fit: problem

When does it happen?

- The peak finder find two nearby subevent peaks. Their time windows overlaps $\rightarrow$ merged into one larger window.
- A delayed Michel electron may therefore be inside the same gate as the primary particle.
- In one subevent window a PMT contributes at most one selected hit: if primary-particle light and decay-electron light are in the same window, they contaminate each other if hits are not separated .

Association criterion

- Define a smaller time window inside the subevent to distinguish the two particles:

$$T^{i}_{\mathrm{res},k} = t^{\mathrm{raw}}_{i} - t^{(k)}_{\mathrm{peak}} - \frac{|\boldsymbol{R}_i - \boldsymbol{x}^{\mathrm{pre}}_{\mathrm{vtx}}|}{c_n}, \quad -30\ \mathrm{ns} < T^{i}_{\mathrm{res},k} < 60\ \mathrm{ns}$$

with $t^{(k)}_{\mathrm{peak}}$ the time peaks previously founded by peak finder alg.

10. In-gate decay-electron fit: hit allocation

Reconstruction rule

- For each in-gate decay-electron subevent except the primary (i.e. the first one), PMTs satisfying the cut are treated as hit.
- PMTs not satisfying the cut are treated as unhit for that subevent.
- The primary subevent is reconstructed after masking hits associated with the other in-gate subevents.

Meaning

- The same merged time window is decomposed into separate hit maps.
- Each hit map is then passed to the ordinary single-ring $e/\mu/\pi^+$ fitter.

Efficiency

$50\ \mathrm{cm} < D_{\mathrm{wall}} < 200\ \mathrm{cm}$		
Sample	Type	fiTQun
Sub-GeV Muon	Data	$89.47\% \pm 0.08\%$
	MC	$88.72\% \pm 0.15\%$
Multi-GeV Muon	Data	$61.98\% \pm 0.12\%$
	MC	$59.91\% \pm 0.23\%$

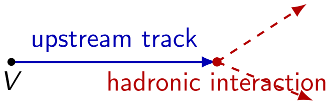
11. Upstream-track π^+ fit: physical motivation

Charged pion in water

- A π^+ can produce Cherenkov light while propagating.
- It can then undergo hadronic scattering or absorption on nuclei.
- After the hadronic interaction, the direction and light pattern may no longer look like one clean track.

Target topology

thin upstream ring = Cherenkov light before the hadronic interaction



downstream light
not modeled as one
regular track

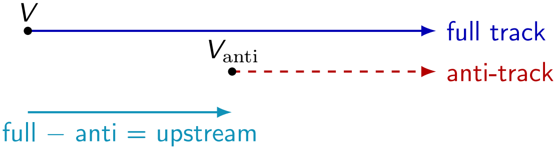
Why not a normal full track?

- A full-track hypothesis would predict light from the downstream part too.
- The upstream-track hypothesis predicts only the clean truncated segment.

11. Upstream-track π^+ fit: predicted charge

Construction

$$\boxed{\mu_i^{\pi^+,\mathrm{up}} = \mu_i^{\mathrm{full}}(p, V, \hat{\boldsymbol{d}}) - \mu_i^{\mathrm{anti}}(p_{\mathrm{anti}}, V_{\mathrm{anti}}, \hat{\boldsymbol{d}})}$$



Meaning

- The anti-track is not a real particle.
- It subtracts the part of the full-track light that would come after the hadronic interaction.
- The same PMT likelihood is used, but with $\mu_i^{\pi^+,\mathrm{up}}$.

11. Upstream-track π^+ fit: role of $E_{\rm loss}$

Fit parameters

$$X_{\pi^+}^{\rm up} = \{t, x, y, z, \theta, \phi, p, E_{\rm loss}\}$$

$E_{\rm loss}$ defines the truncation point

$$L_{\rm up} \simeq \frac{E_{\rm loss}}{dE/dx}, \qquad V_{\rm anti} = V + L_{\rm up}\hat{\boldsymbol{d}}$$

Residual momentum of the anti-track

$$K_{\rm init} = \sqrt{p^2 + m_\pi^2} - m_\pi, \qquad K_{\rm anti} = K_{\rm init} - E_{\rm loss}$$

$$p_{\rm anti} = \sqrt{(K_{\rm anti} + m_\pi)^2 - m_\pi^2}$$

Important interpretation

- $E_{\rm loss}$ is not measured separately by a PMT.
- It is a free fit parameter chosen together with $V, t, \hat{\boldsymbol{d}}, p$.
- For timing, the midpoint of the upstream segment is used, not the full track.

11. μ/π^+ identification variables

For each event, run two fits

$$L_\mu^{\mathrm{best}} = L_\mu(\hat{X}_\mu), \qquad L_{\pi^+}^{\mathrm{best}} = L_{\pi^+}(\hat{X}_{\pi^+}^{\mathrm{up}})$$

Useful variables

$$\ln \frac{L_{\pi^+}}{L_\mu}, \qquad \widehat{E}_{\mathrm{loss}}$$

Why a 2D separation?

- Muons and π^+ are both non-showering and have similar Cherenkov patterns.
- A true muon can often be fitted almost equally well by a π^+ upstream hypothesis.
- The fitted E_{loss} tells how the pion hypothesis had to truncate the track.

Practical use

- MC separation in $\widehat{E}_{\mathrm{loss}}$ vs. $\ln(L_{\pi^+}/L_\mu)$.
- The cut is not universal: it must be optimized for the analysis sample.

12. π^0 fit: physical hypothesis

Background problem

- A neutral pion decays as

$$\pi^0 \to \gamma\gamma.$$

- Each photon produces an electromagnetic shower and therefore an e-like ring.
- If the photons overlap, a real π^0 can look like one e-like ring.

Two-photon hypothesis

$$\mathcal{H}_{\pi^0} = \{\gamma_1, \gamma_2 \text{ from a common } \pi^0 \text{ vertex/time}\}$$

Fit outputs

- photon momenta: $p_{\gamma 1}, p_{\gamma 2}$;
- photon directions: $\hat{\boldsymbol{d}}_1, \hat{\boldsymbol{d}}_2$;
- conversion lengths: $d_{\mathrm{conv},1}, d_{\mathrm{conv},2}$;
- common π^0 vertex and creation time.

12. π^0 fit: likelihood comparison

Same PMT data, different hypotheses

$$\mathcal{H}_e : \text{one electromagnetic ring}$$

$$\mathcal{H}_{\pi^0} : \text{two electromagnetic rings from } \gamma\gamma$$

Predicted charge for two photons

$$\mu_i^{\pi^0} = \mu_{i,\gamma 1} + \mu_{i,\gamma 2}$$

Reconstructed invariant mass

$$m_{\gamma\gamma}^2 = 2E_1 E_2 \left(1 - \cos\theta_{\gamma\gamma}\right)$$

e/π^0 **discrimination**

- Use $\ln(L_{\pi^0}/L_e)$ and reconstructed $m_{\gamma\gamma}$.
- For a true electron, $m_{\gamma\gamma}$ is only the result of forcing a false two-photon hypothesis.
- For a true π^0, the fit tends to recover a mass near m_{π^0} when both photons are reconstructible.

13. Multi-ring fit: what is being fitted?

Applied to the primary subevent

- Multi-ring reconstruction targets prompt particles from the primary interaction.
- Delayed decay-electron subevents are handled by subeventing / in-gate masking.

Multi-ring hypothesis

$$\mathcal{H}_{N\mathrm{R}} = \{N \text{ simultaneous Cherenkov rings in the same subevent}\}$$

For each ring k

$$X_k = \{t_k, \boldsymbol{x}_k, \hat{\boldsymbol{d}}_k, p_k, \mathrm{PID}_k\}$$

Charge prediction for each PMT

$$\mu_i^{\mathrm{tot}} = \sum_{k=1}^{N} \mu_{i,k}$$

- PMTs are not hard-assigned to a ring in the final likelihood.
- A PMT in an overlap region can receive light from several rings.

13. Multi-ring time likelihood

For each PMT and each ring

$$t^{(k)}_{\mathrm{res},i} = t^{\mathrm{raw}}_i - t_k - t^{(k)}_{\mathrm{prop}}$$

One PMT records one hit time

- The registered time is controlled by the first detected photon.
- Residual times are not averaged over rings.

Conceptual first-arrival pdf for two sources

$$f_{\mathrm{tot}}(t) = f_1(t)\,[1 - F_2(t)] + f_2(t)\,[1 - F_1(t)]$$

fiTQun approximation

- Time PDFs are computed ring by ring.
- They are merged assuming photons from the ring with earlier expected time t^{exp}_i arrive before photons from rings with later t^{exp}_i.
- Predicted charge matters: more expected photons make earlier first hits more likely.

13. Initial multi-ring fit

Basic strategy

1. Start from the best lower-ring hypothesis.
2. Search for an additional ring direction using the remaining ring-like structure.
3. Try two broad PID classes for the newly added ring:

$$e\text{-like}, \qquad \pi^+\text{-like}.$$

4. Refit the event with the enlarged multi-ring hypothesis.
5. Apply fake-ring reduction before accepting the extra ring.

Why only e and π^+ initially?

- e-like represents showering rings: $e^\pm$ and γ.
- π^+-like represents non-showering rings: $\mu^\pm$ and charged pions.
- This keeps the combinatorics manageable while preserving the main ring topologies.

13. Ring counting vs. physical particles

Reconstructed ring count is not truth multiplicity

$$N_{\mathrm{ring}}^{\mathrm{reco}} \neq N_{\mathrm{particles}}^{\mathrm{true}}$$

Examples

- True $\pi^0 \to \gamma\gamma$ can be reconstructed as one ring if the photons overlap.
- A showering electron can create fake substructure and seed a fake ring.
- A proton or kaon above threshold may be absorbed by the nearest available non-showering hypothesis.

What the fitter really asks

Does adding another Cherenkov source improve the PMT charge-time pattern enough?

Why fake-ring reduction is needed

- More rings mean more free parameters.
- More parameters almost always improve the raw likelihood.
- A likelihood improvement must be large and physically meaningful to keep the extra ring.

14. Sequential multi-ring fit: motivation

Problem after the initial multi-ring fit

- Some rings can be fake.
- Some rings can have wrong PID.
- Some nearby rings can be fragments of the same physical ring.

Sequential refit idea

1. Reorder rings by visible energy.
2. Start from the most energetic ring.
3. Merge lower-energy rings within 20° into it by adding visible energy.
4. Refit that ring while keeping the other rings fixed.
5. Repeat in descending visible energy.

Meaning

- This refines PID and kinematics after the rough initial ring-counting stage.
- The refit is effectively a one-ring fit in the presence of fixed additional rings.

14. Sequential multi-ring fit: PID rule

For the ring being refitted

- Test electron, muon and upstream-track π^+ hypotheses.
- The showering / non-showering decision is made with

$$\ln \frac{L_e}{L_{\pi^+}} > -10 \quad \Rightarrow \quad e\text{-like ring.}$$

Current practical rule

- The dedicated μ/π^+ identification is not applied in the sequential fit.
- If the most energetic ring is non-showering, it is assumed to be a muon.
- Other non-showering rings are assumed to be π^+.

Physical reason

- A most-energetic sharp ring usually indicates a ν_μ CC event with a true muon.
- A lower-energy sharp ring is more likely to be a charged pion from the hadronic system.

14. Multi-ring output and interpretation

Final multi-ring result

- Number of rings kept after fake-ring reduction and sequential refit.
- For each ring: PID class, momentum, direction, time and vertex.
- Ring vertices need not remain exactly identical after sequential refitting.

Interpretation rules

- Multi-ring means multiple simultaneous Cherenkov sources in the primary subevent.
- It does not include delayed Michel electrons as prompt rings.
- It is a reconstruction hypothesis, not a direct list of MC truth particles.

Subeventing vs Multi-ring

subeventing separates time clusters, multi-ring separates topology inside one time cluster.

15. What is on-the-fly and what is precomputed?

Object	**Precomputed / tuned**	**On-the-fly in fit**
Vertex goodness $G(\boldsymbol{x}, t)$	no MC table; formula fixed	evaluate using selected PMT times
Peak threshold $F(t)$	formula and constants η, γ fixed	compute from local maxima of current event scan
Cherenkov profile g	particle-gun MC; profile tables / integrals	interpolate for current geometry and momentum
Direct charge μ^{dir}	integrals I_n, angular response, optical parameters	compute geometry, j_n, interpolate integrals
Scattered charge μ^{sct}	scattering table $A(s)$ from WCSim MC	interpolate $A(s)$, compute k_n, interpolate integrals
Unhit probability	polynomial coefficients from simulation/tuning	evaluate at $\mu_i(X)$
Charge PDF f_q	charge-response MC and fitted functions	evaluate at measured q_i and $\mu_i(X)$
Time PDF f_t	direct/indirect time PDF parameters from MC/tuning	compute t_{res}, evaluate at $\mu_i(X)$, merge direct/scattered PDFs

16. Chronological summary

Input event: PMT charges q_i, raw times t_i^{raw}, PMT geometry $\boldsymbol{R}_i$.

Prompt vertex pre-fit: use hits in $[-100, +400]$ ns; maximize $G(\boldsymbol{x}, t)$; obtain rough $\boldsymbol{x}_{\mathrm{vtx}}^{\mathrm{pre}}$.

Peak finder: fix $\boldsymbol{x}_{\mathrm{vtx}}^{\mathrm{pre}}$; scan $G(t)$ every 8 ns using hits in $[-200, 15000]$ ns.

Peak selection: local maxima M; dynamic threshold $F(t)$; first peak above F; later peaks require valley below $0.6F$.

Time windows: for each peak, select hits with $-180 < T_{\mathrm{res}} < 800$ ns; window edges from min/max raw hit times; merge overlaps.

Single-ring fit per subevent: for each particle hypothesis a, seed X; minimize $-\ln L_a(X)$.

Special hypotheses: in-gate hit masking, upstream-track π^+, $\pi^0 \to \gamma\gamma$, and multi-ring fits for the primary subevent.

Likelihood evaluation: compute predicted PMT charges/times; compare single-ring and multi-ring hypotheses.

Output: best-fit vertex, time, direction, momentum, PID/topology; decay-electron and multi-ring variables.